



CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No. -01

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Recap of Previous Lecture



Topic

Magnetic Disk ✓

Topic

Disk Capacity

Topic

Disk Access Time

Topics to be Covered



Topic

Disk Addressing

Topic

Pipeline Processing

Topic

Speed Up

Ques) A file is stored on 5 consecutive tracks on all their sectors

$$\text{No. of sectors/track} = 200$$

$$\text{no. of sectors} = 5 * 200 = 1000$$

$$\text{Disk rotat}^n \Rightarrow 6000 \text{ rpm} \rightarrow 1 \text{ rotat}^n \text{ time} = 10 \text{ ms}$$

$$\text{seek time} = 10 \text{ msec}$$

$$\text{file access time} = \text{_____ msec?}$$

$$= (5 * 10) + 5 * \frac{10}{2} + 1000 * \frac{10}{200}$$

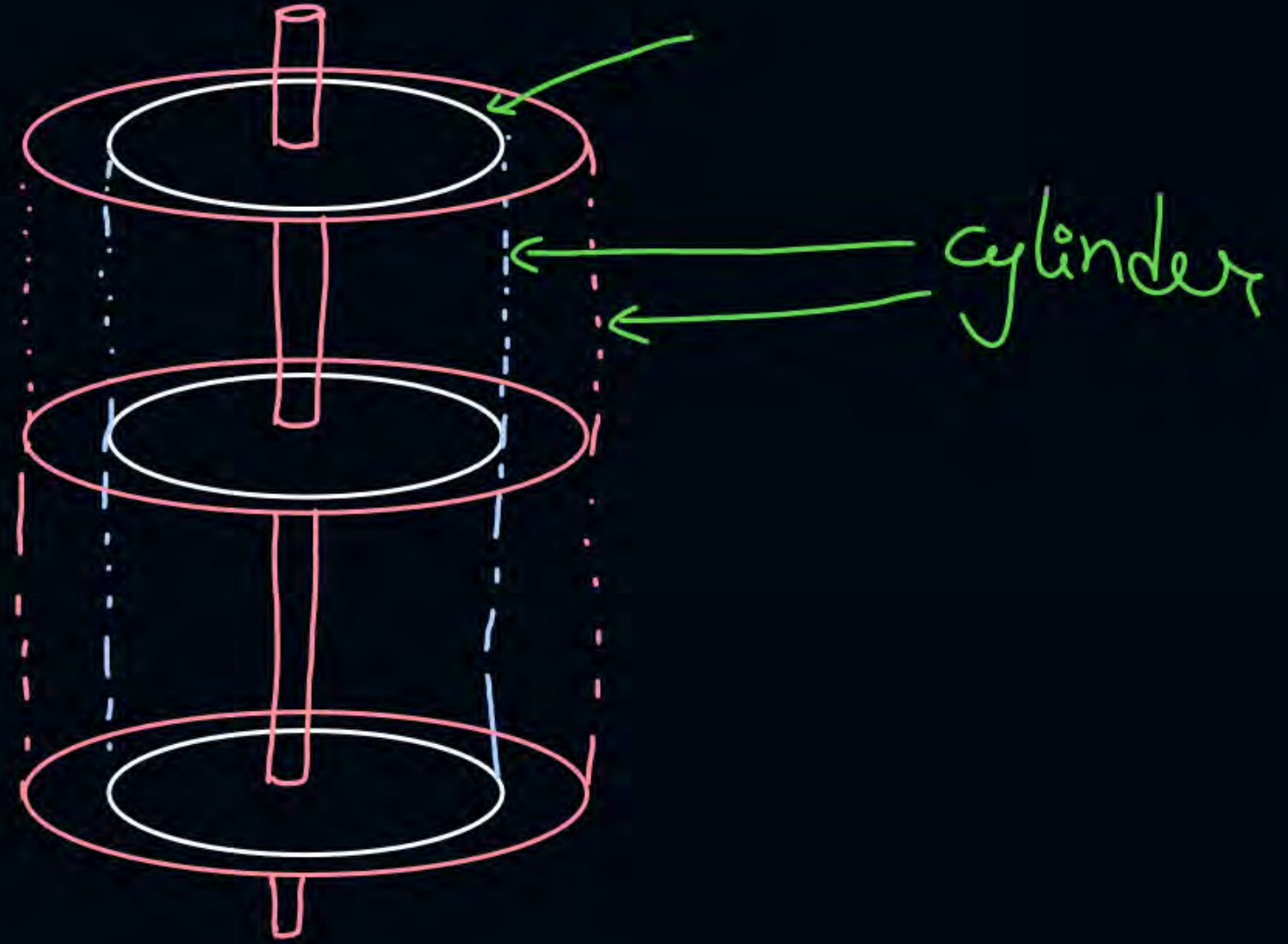
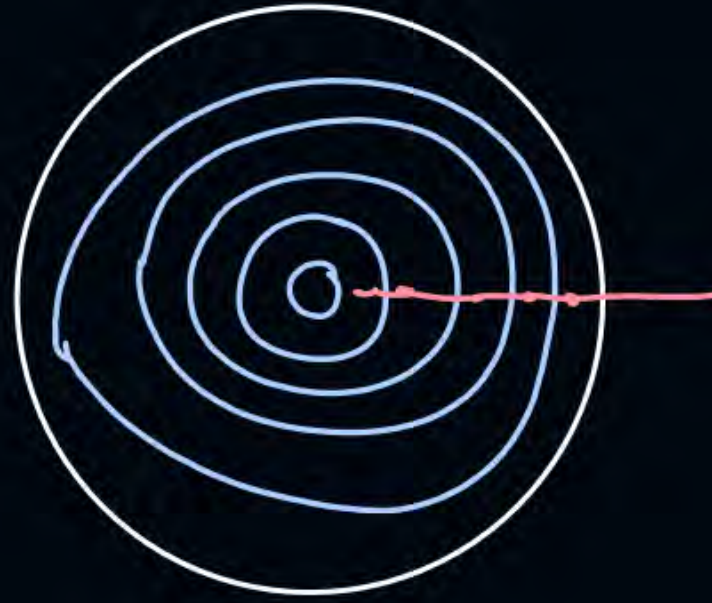
$$= 50 + 25 + 50$$

$$= 125 \text{ msec}$$

Assume all 5 tracks are under a cylinder in last queue,
then

$$\begin{aligned}\text{file access time} &= 10 + 5 * \frac{10}{2} + 1000 * \frac{10}{200} \\ &= 85 \text{ ms}\end{aligned}$$

cylinder :-

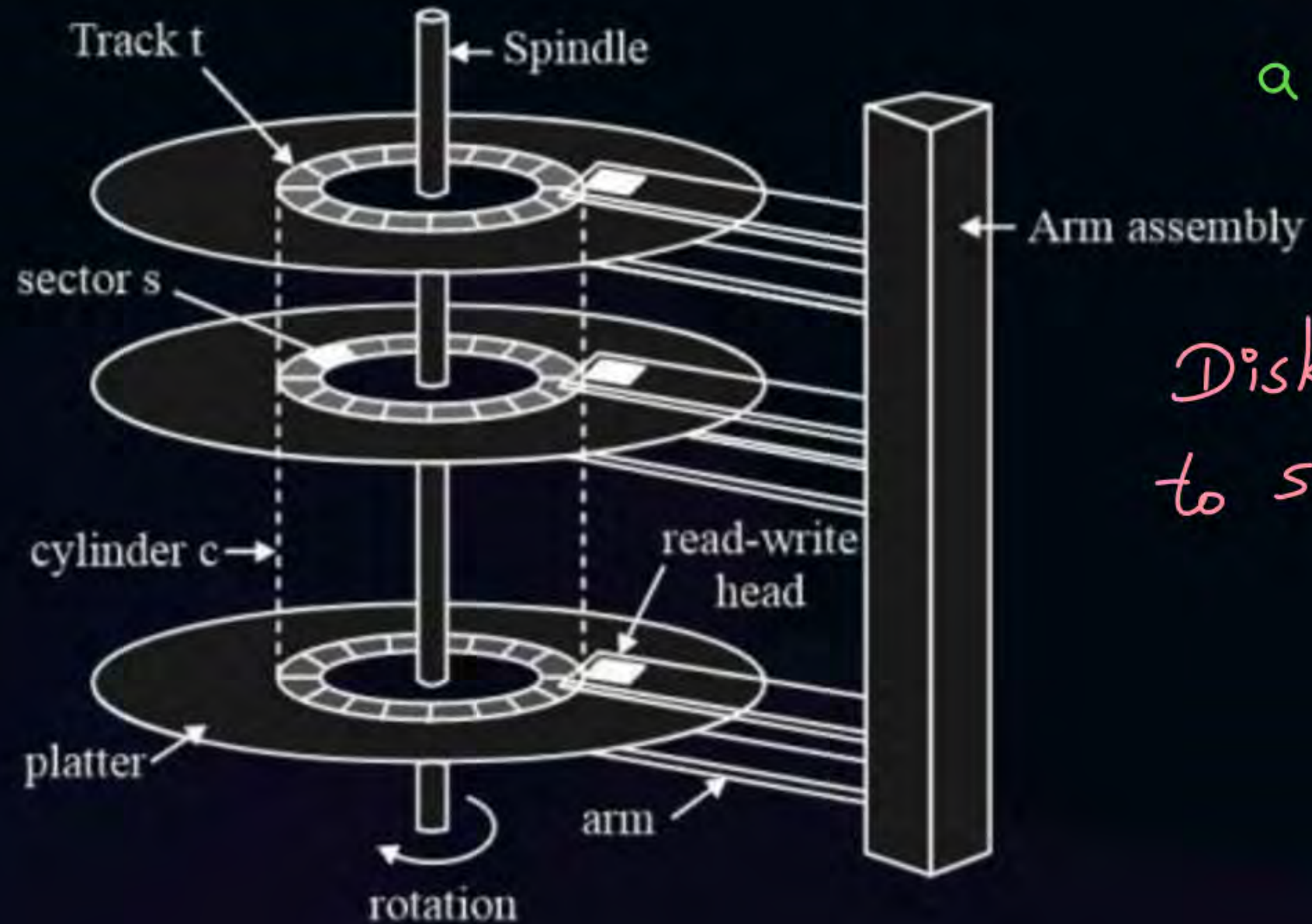




Topic : Cylinder



collection of same radius tracks from all surfaces, form a cylinder.



Disk is accessed cylinder wise to save seek time!



Topic : Disk Addressing



$\langle \text{cylinder no, surface no, sector no} \rangle$

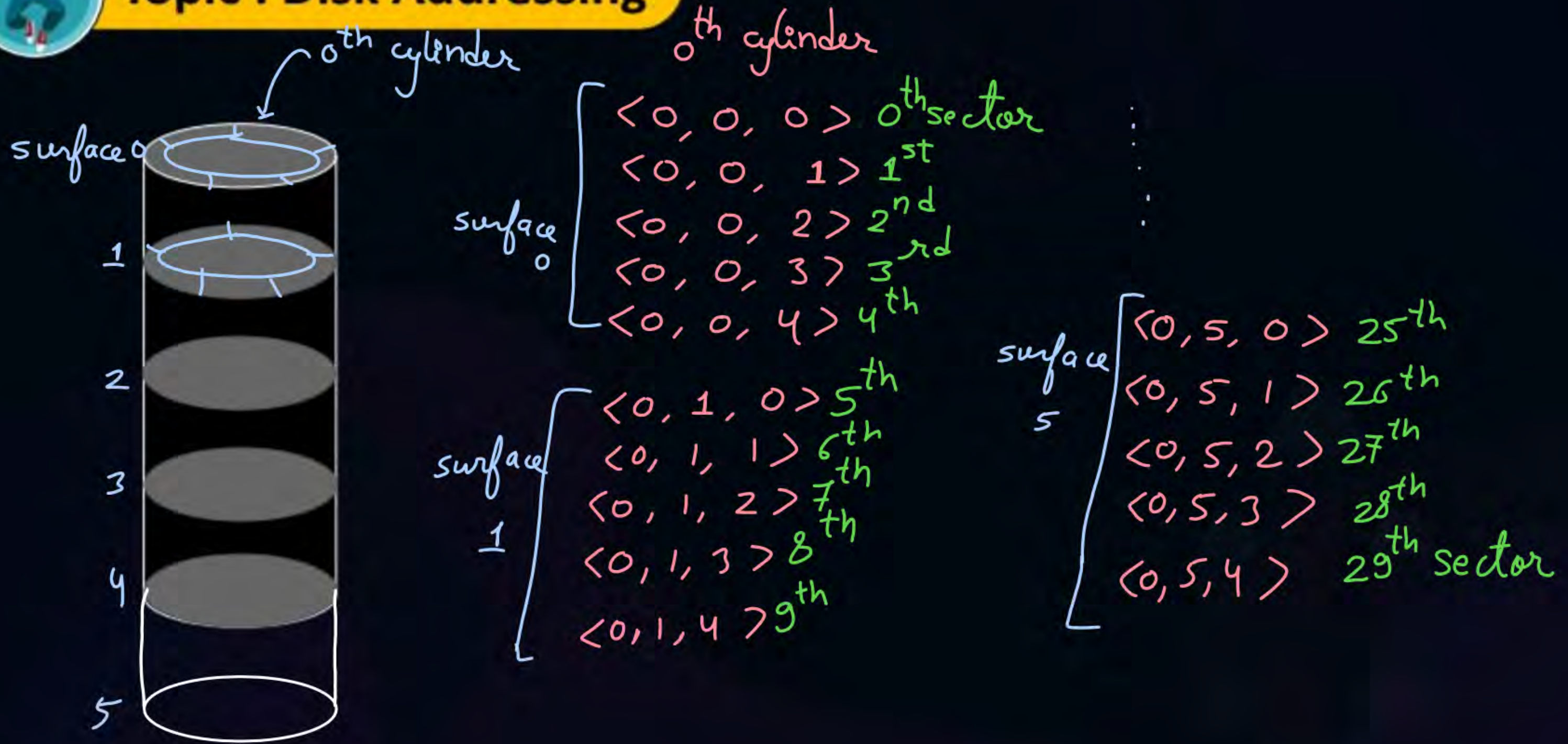
Assume

- Disk has 3 platters with 2 recording surfaces on each $\Rightarrow$ no. of surfaces in disk = 6
- Number of tracks per surface = 4
- Number of sectors per track = 5

$$\begin{aligned}\text{no. of sectors per cylinder} &= \text{no. of surfaces} * \text{no. of sectors per track} \\ &= 6 * 5 \\ &= 30\end{aligned}$$



Topic : Disk Addressing



cylinder 1

Surface 0 $\left[\begin{array}{l} \langle 1, 0, 0 \rangle \\ \langle 1, 0, 1 \rangle \\ \langle 1, 0, 2 \rangle \\ \langle 1, 0, 3 \rangle \\ \langle 1, 0, 4 \rangle \end{array} \right.$

30th sector

31st

32nd

33rd

34th

⋮

53rd 1,

Surface 5 $\left[\begin{array}{l} \langle 1, 5, 0 \rangle \\ \vdots \\ \langle 1, 5, 4 \rangle \end{array} \right.$

55th sector

59th sector

Address:- $\langle 1, 5, 2 \rangle$

belongs to which sector no?

no. of sectors before cylinder 1 = 30
(1 complete cylinder)

no. of sectors covered before surface 5 = 5×5
in cylinder 1 = 25
(5 complete surfaces 0 to 4)

sector of surface $\frac{= 2}{57}$

Address $\Rightarrow \langle c, h, s \rangle$

cylinder no.

surface no.

sector no.

Add. belongs to sector = $(c * n_c) + (h * n_t) + s$

$$n_c = \text{no. of sectors per cylinder} = \text{no. of surfaces in disk} * n_t$$

$$n_t = -11 \text{ ————— per track}$$

$$c = \lfloor \text{sector no.} / n_c \rfloor$$

$$h = \lfloor (\text{sector no.} \% n_c) / n_t \rfloor$$

$$s = (\text{sector no.} \% n_c) \% n_t$$

53rd sector

$$c = \lfloor 53 / 30 \rfloor = 1$$

$$h = \lfloor (53 \% 30) / 5 \rfloor = 4$$

$$s = (53 \% 30) \% 5 = 3$$

$$\text{add.} = \langle 1, 4, 3 \rangle$$

$$\text{Ans} = 1644$$

#Q. A hard disk has 16 sectors per track, 4 platters each with 2 recording surfaces and 32 cylinders. The address of a sector is given as a triple $\langle c, h, s \rangle$, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as $\langle 0, 0, 0 \rangle$, the 1st sector as $\langle 0, 0, 1 \rangle$, and so on.

The address $\langle 12, 6, 12 \rangle$ corresponds to sector number?

$$n_t = 16$$

$$n_c = 8 * 16 = 128$$

$$\begin{aligned} \text{sector no.} &= (12 * 128) + (6 * 16) + 12 \\ &= \underline{\underline{1644}} \end{aligned}$$

$$c = \lfloor 1644 / 128 \rfloor = 12$$

$$h = \lfloor (1644 \% 128) / 16 \rfloor = 6$$

$$s = (1644 \% 128) \% 16 = 12$$

$$\text{add.} = \langle 12, 6, 12 \rangle$$

#Q. A hard disk has 16 sectors per track, 4 platters each with 2 recording surfaces and 32 cylinders. The address of a sector is given as a triple $\langle c, h, s \rangle$, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as $\langle 0, 0, 0 \rangle$, the 1st sector as $\langle 0, 0, 1 \rangle$, and so on.

Ans $\langle 6, 6, 3 \rangle$

The address of 867th sector?

$$n_t = 16$$
$$n_c = 8 * 16 = 128$$

$$c = \lfloor 867 / 128 \rfloor = 6$$
$$h = \lfloor (867 \% 128) / 16 \rfloor = 6$$
$$s = (867 \% 128) \% 16 = 3$$

#Q. Consider a hard disk with 36 recording surfaces (0-35) having 10000 cylinders (0-9999) and each track contains 64 sectors (0-63). Data in disk are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file in the disk is stored starting from address <1660, 28, 38>. What is the sector number of the first sector of the file in the disk?

$$n_t = 64$$
$$n_c = 36 * 64 = 2304$$

$$\begin{aligned} \text{sector no. of first sector the file} &= (1660 * 2304) \\ &+ 28 * 64 \\ &+ 38 \\ \hline &38,26,470 \end{aligned}$$

#Q. In above questions if a file is stored on 55788 sectors in contiguous manner then what is the sector number of the last sector of the file?

$$\begin{array}{r} = 38\ 26\ 470 \\ + \quad 55\ 788 \\ - 1 \\ \hline 38\ 82\ 257 \end{array}$$

[NAT]



from last Questⁿ

#Q. Calculate the address in format $\langle c, h, s \rangle$ for the last sector of the file?

$$c = \lfloor 3882257 / 2304 \rfloor = 1685$$

$$\langle 1685, 0, 17 \rangle$$

$$h = \lfloor (3882257 \% 2304) / 64 \rfloor = 0$$

$$s = (3882257 \% 2304) \% 64 = 17$$

$$\text{Ans} = \underline{\underline{1284}}$$

#Q. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each track contains 64 sectors (0-63). Data storage capacity of in each sector is 512 Bytes. Data are organized cylinder-wise and addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

$$n_t = 64$$

$$n_c = 16 * 64 = 1024$$

$$\begin{aligned} \text{sector no. of first sector of file} &= (1200 * 1024) + (9 * 64) + 40 \\ &= 1229416 \end{aligned}$$

$$\text{no. of sectors needed to store file} = \frac{42797 \overset{2}{\cancel{\text{KB}}}}{\cancel{512} \cancel{8}} = 85594$$

$$\text{sector no. of last sector of file} = \begin{array}{r} 1229416 \\ + 85594 \\ - 1 \\ \hline 1315009 \end{array}$$

$$c = \lfloor 1315009 / 1024 \rfloor = 1284$$

$$\langle 1284, 3, 1 \rangle$$

$$h = \lfloor (1315009 \% 1024) / 64 \rfloor = 3$$

$$s = (1315009 \% 1024) \% 64 = 1$$



Topic : Parallel Processing

Parallel Processing: Simultaneous data processing

Types:

- Vector Processing
- Array Processing
- ✓ Pipeline Processing (Pipelining)



Topic : Flynn's Classification of Computers

1. SISD (single Instⁿ stream single Data stream)
2. SIMD ——— || ——— multiple — || ——— $\Rightarrow$ Pipeline processor
3. MISD multiple — || ——— single — || ——— $\Rightarrow$ not practical
4. MIMD multiple — || ——— multiple — || ——— $\Rightarrow$ super scalar computers
(multiple pipelines)



Topic : Pipeline Processing

- Pipelining is useful, When same processing is applied over multiple inputs



Topic : Pipeline Processing

- Technique to decompose a sequential process into sub-operations
- Sub-operations are performed in **segments** or *stages*.
- **Task:** One operation performed in all segments

↳ *no. of inputs to be processed.*



Topic : Pipelining Example

$$(4 + 5 - 1) = 8 \text{ cycles}$$



cycles		Person 1 (cut)	Person 2 (stitch)	Person 3 (finish)	Person 4 (Pack)
1	10 min	cut ₁			
2	20 min	cut ₂	stitch ₁		
3	30 min	cut ₃	stitch ₂	finish ₁	
4	40 min	cut ₄	stitch ₃	finish ₂	Pack ₁
5	50 min	cut ₅	stitch ₄	finish ₃	Pack ₂
6	60 min	—	stitch ₅	finish ₄	Pack ₃
7	70 min	—	—	finish ₅	Pack ₄
8	80 min	—	—	—	Pack ₅



Topic : Pipeline Cycle Time

(t_p)



Min. time in which all segments can finish their respective sub-operations.





Topic : General Consideration About Pipeline

- Consider a k segment pipeline with clock cycle time = t_p to perform n tasks

$$\text{time needed to perform 1}^{\text{st}} \text{ task} = \overset{\text{no. of cycles}}{k} \quad \overset{\text{time}}{k * t_p}$$

$$\text{time needed to perform remaining } (n-1) \text{ tasks} = (n-1) \quad (n-1) * t_p$$

$$\text{Total time needed to perform all } n \text{ tasks} = (k+n-1) \quad (k+n-1) t_p$$



Topic : General Consideration About Pipeline

- Consider a non-pipeline system that takes t_n time to perform a task

time needed to perform n tasks in non-pipeline system = $n * t_n$



Topic : General Consideration About Pipeline

- Performance of a pipeline is given by **Speed up ratio**.

$$\text{Speed up } (S) = \frac{\text{non-pipeline time}}{\text{pipeline time}}$$

$$S = \frac{n * t_n}{(k+n-1) t_p}$$

when no. of tasks increases
 $n \gg k$

$$S_{\text{ideal}} = \frac{t_n}{t_p}$$

max speed up
possible

★ Special case:-

if one operation takes same time in
pipeline & non-pipeline both. then

$$t_n = k * t_p$$

$$S_{\text{ideal}} = k$$



2 mins Summary



Topic

Disk Addressing

Topic

Pipeline Processing

Topic

Speed Up



Happy Learning

THANK - YOU